Homogeneity and universality for countable coherence spaces

Abstract

Kou and Zhao constructed in 2015 a countable universal coherence space C_ω from finite partial binary functions. We determine its homogeneity status. The empty partial function is the unique event having no coherent neighbour distinct from itself, and this invariant shows that C_ω is not ultrahomogeneous. For comparison, we recall the classical arithmetic Rado graph, regarded as a reflexive coherence space. This direct binary-digit presentation goes back to Rado, and universal homogeneous objects in coherence spaces and qualitative domains were studied explicitly by Boldi, Cardone, and Droste. We include self-contained proofs of the extension property, countable universality, ultrahomogeneity, and uniqueness in order to make the contrast with C_ω precise. The appendices develop the application to the standard coherence-space semantics of propositional linear logic. Every countable formula denotation is a linear retract of $D(R_{\mathrm{coh}})$, typed denotations are represented by endomaps satisfying a Karoubi idempotent equation, and the state domain of the original coordinate construction admits an explicit antichain and ideal-completion description.

Keywords: coherence space, Rado graph, ultrahomogeneity, universal structure, linear logic, retract
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1. Introduction

A coherence space is a reflexive undirected graph whose states are its cliques. Coherence spaces form one of the original denotational models of linear logic [4, 5]. Their graph presentation also makes questions about universal and homogeneous structures available to direct combinatorial analysis.

In 2015, Hui Kou and Haoran Zhao constructed a universal coherence space from the compact elements of $\mathbb{T}^\omega$ [7]. In the equivalent finite-coordinate presentation, its events are finite partial functions from $\mathbb{N}$ to $\{0, 1\}$. Distinct events are coherent when they assign different values at a common coordinate. We denote this space by C_ω. The original construction proves that every at-most-countable coherence space occurs as an induced subspace of C_ω; on state domains, this is exactly the mechanism that produces a linear retraction. The source does not analyse ultrahomogeneity; it does not, however, formulate that issue as an explicit open problem.

Our main observation determines this additional property of the coordinate construction. The empty partial function is fixed by every automorphism of C_ω, since it is the unique event coherent only with itself. Nevertheless, every one-point induced subspace is isomorphic to every other one. Mapping the empty function to any nonempty function therefore gives a finite partial isomorphism that cannot extend to an automorphism.

For comparison, we use the standard arithmetic presentation of the Rado graph [3, 8] and add a loop at every vertex, as required by the coherence-space convention. The definition is direct rather than recursive: adjacency of two natural numbers is read from a binary digit of the larger one. The graph itself and this arithmetic presentation are classical; the recursive step enters only in the proof that every countable graph embeds. The resulting coherence space R_{coh} satisfies the finite extension property: for every pair of disjoint finite sets of events, a fresh event can be chosen coherent with all events in the first set and incoherent with all events in the second. The usual recursive and back-and-forth arguments then imply countable universality, ultrahomogeneity, and uniqueness up to isomorphism. More generally, explicit universal homogeneous objects for coherence spaces and qualitative domains were already constructed by Boldi, Cardone, and Droste [2]. Thus the role of R_{coh} here is a classical comparison object, not a new homogeneous construction.

The distinction between the two structures is therefore precise. The finite-partial-function construction remains an explicit universal coding, but it has a definable exceptional event. The Rado construction realizes every finite coherence pattern and has no such exceptional finite behaviour.

Appendix A recalls the syntax, linear traces, categorical operations, and sound interpretation of propositional linear logic. Appendix B then applies the homogeneous universal space. Writing $U = D(R_{\mathrm{coh}})$, every countable

formula interpretation is a linear retract of U. After choosing a section–retraction pair for each formula, the formula is represented by a linear idempotent on U, while a typed denotation from A to B is represented by an endomap g satisfying $g = e_B g e_A$. Appendix C records the concrete state domain carried by the original finite-coordinate construction. The choices of embeddings into R_{coh} are not canonical.

2. Coherence spaces, embeddings, and retracts

Definition 2.1. A *coherence space* is a pair $E = (|E|, \smile_E)$ consisting of an event set $|E|$ and a reflexive symmetric relation $\smile_E$ on $|E|$. A *state* of E is a subset $X \subseteq |E|$ such that

$$x \smile_E y \qquad (x, y \in X).$$

The states, ordered by inclusion, form the state domain

$$D(E) = \{X \subseteq |E| : X \text{ is a state of } E\}.$$

All coherence spaces in the sequel are at most countable. The reflexive relation may equivalently be regarded as a simple graph by deleting the loops.

Definition 2.2. Let E and F be coherence spaces. An *induced embedding* $\varphi : E \hookrightarrow F$ is an injective map $\varphi : |E| \to |F|$ such that, for all $x, y \in |E|$,

$$x \smile_E y \quad \Longleftrightarrow \quad \varphi(x) \smile_F \varphi(y).$$

The space F is *countably universal* if every at-most-countable coherence space induced-embeds into F.

Definition 2.3. A coherence space E is *ultrahomogeneous* if every isomorphism between finite induced subspaces of E extends to an automorphism of E.

The connection with state-domain retracts is the following elementary fact.

Proposition 2.4. *If $\varphi : E \hookrightarrow F$ is an induced embedding, then the maps*

$$I_\varphi : D(E) \longrightarrow D(F), \qquad I_\varphi(X) = \varphi[X],$$

and

$$P_\varphi : D(F) \longrightarrow D(E), \qquad P_\varphi(Y) = \{x \in |E| : \varphi(x) \in Y\}$$

are linear stable maps and satisfy

$$P_\varphi I_\varphi = \mathrm{id}_{D(E)}, \qquad I_\varphi P_\varphi \leq \mathrm{id}_{D(F)}.$$

In particular, $D(E)$ is a linear retract of $D(F)$.

Proof. Preservation of coherence makes $I_\varphi(X)$ a state, and reflection of coherence makes $P_\varphi(Y)$ a state. The two maps are represented by the linear traces

$$\{(x, \varphi(x)) : x \in |E|\}, \qquad \{(\varphi(x), x) : x \in |E|\},$$

respectively. Injectivity gives $P_\varphi I_\varphi(X) = X$, while $I_\varphi P_\varphi(Y) = Y \cap \varphi[|E|] \subseteq Y$. □

3. The finite-partial-function universal space

Let

$$\mathcal{P} = \mathrm{Fn}_{<\omega}(\mathbb{N}, 2)$$

be the set of finite partial functions from $\mathbb{N}$ to $\{0, 1\}$. For $s, t \in \mathcal{P}$, write $s \# t$ when they conflict:

$$s \# t \quad \Longleftrightarrow \quad \exists i \in \mathrm{dom}(s) \cap \mathrm{dom}(t),\ s(i) \neq t(i).$$

Definition 3.1. The coherence space $C_\omega = (\mathcal{P}, \smile_\omega)$ is defined by

$$s \smile_\omega t \quad \Longleftrightarrow \quad s = t \text{ or } s \# t.$$

Proposition 3.2. *The coherence space C_ω is countably universal. Consequently every state domain of an at-most-countable coherence space is a linear retract of $D(C_\omega)$.*

Proof. Let E be an at-most-countable coherence space and fix an injection $c : |E| \to \mathbb{N}$. For $x \in |E|$, define $\varphi(x) \in \mathcal{P}$ by

$$\varphi(x)(c(x)) = 1$$

and, whenever $c(y) < c(x)$,

$$\varphi(x)(c(y)) = 0 \quad \Longleftrightarrow \quad y \smile_E x.$$

All remaining coordinates are undefined. The domain is finite because all its coordinates are at most $c(x)$. The map φ is injective: its distinguished value 1 occurs at $c(x)$.

Suppose $c(x) < c(y)$. The only coordinate at which $\varphi(x)$ and $\varphi(y)$ can have different defined values is $c(x)$. They conflict there exactly when $x \smile_E y$. Thus

$$x \smile_E y \quad \Longleftrightarrow \quad \varphi(x) \smile_\omega \varphi(y),$$

so φ is an induced embedding. The retract statement follows from Proposition 2.4. □

Universality does not imply homogeneity for this construction.

Theorem 3.3. *The coherence space C_ω is not ultrahomogeneous.*

Proof. Let ε be the empty partial function. It cannot conflict with any event, hence

$$\varepsilon \smile_\omega s \quad \Longleftrightarrow \quad s = \varepsilon.$$

Conversely, if $s \neq \varepsilon$, choose $i \in \mathrm{dom}(s)$ and let t be the one-coordinate partial function with $t(i) = 1 - s(i)$. Then $s \# t$, so s has a coherent neighbour distinct from itself. Therefore ε is the unique event coherent only with itself, and every automorphism of C_ω fixes it.

For any nonempty s, the one-point induced subspaces $\{\varepsilon\}$ and $\{s\}$ are isomorphic. The isomorphism sending ε to s cannot extend to an automorphism, because every automorphism fixes ε. □

4. The Rado coherence space

For $m < n$, let $\mathrm{bit}_m(n)$ denote the m-th binary digit of n, with the least significant digit numbered 0.

Definition 4.1. The Rado coherence space $R_{\mathrm{coh}} = (\mathbb{N}, \smile_R)$ is defined by reflexivity, symmetry, and the rule

$$m \smile_R n \quad \Longleftrightarrow \quad \mathrm{bit}_m(n) = 1 \qquad (m < n).$$

Definition 4.2. A coherence space R has the *Rado extension property* if, for every pair of disjoint finite sets $A, B \subseteq |R|$, there is $z \in |R| \setminus (A \cup B)$ such that

$$z \smile_R a \quad (a \in A), \qquad z \not\smile_R b \quad (b \in B).$$

Proposition 4.3. *The coherence space R_{coh} has the Rado extension property.*

Proof. Let $A, B \subseteq \mathbb{N}$ be finite and disjoint. Choose N strictly larger than every member of $A \cup B$, and put

$$z = 2^N + \sum_{a \in A} 2^a.$$

The binary digits occurring in this sum are distinct. Moreover $z > N$, so $z \notin A \cup B$. Its a-th binary digit is 1 for $a \in A$, whereas its b-th binary digit is 0 for $b \in B$. Hence z realizes the required coherence and incoherence pattern. □

Theorem 4.4. *Let R be a countable coherence space with the Rado extension property. Then R is countably universal and ultrahomogeneous.*

Proof. Let $E = \{e_0, e_1, \ldots\}$ be an at-most-countable coherence space. Construct an induced embedding recursively. After mapping $e_0, \ldots, e_{n-1}$ to distinct events $r_0, \ldots, r_{n-1}$, apply the extension property to

$$A_n = \{r_i : i < n,\ e_i \smile_E e_n\}, \qquad B_n = \{r_i : i < n,\ e_i \not\smile_E e_n\}.$$

The resulting fresh event r_n has exactly the required relation to the previous image. This recursion proves countable universality.

For ultrahomogeneity, start with an isomorphism between finite induced subspaces of R and enumerate the remaining events. Alternate forth and back steps. At a forth step, use the extension property to map the first event not yet in the domain to a fresh event having the corresponding finite coherence pattern over the current range. A back step applies the same construction to the inverse partial isomorphism. The increasing union is a bijection of R that preserves and reflects coherence, hence an automorphism extending the given finite isomorphism. □

Proposition 4.5. *If a countable coherence space is finitely universal and ultrahomogeneous, then it has the Rado extension property.*

Proof. Let A and B be finite and disjoint. Add to the induced subspace on $A \cup B$ a new event q coherent with every event of A and incoherent with every event of B. Finite universality embeds this finite extension into the given space. The restriction of the embedding to $A \cup B$ is an isomorphism between finite induced subspaces, so ultrahomogeneity extends it to an automorphism. Applying the inverse automorphism to the image of q gives the required event over the original sets A and B. □

Corollary 4.6. *The coherence space R_{coh} is countably universal and ultrahomogeneous.*

Proof. Combine Proposition 4.3 with Theorem 4.4. □

Theorem 4.7. *Any two countable coherence spaces with the Rado extension property are isomorphic. Consequently R_{coh} is unique up to isomorphism among countable, countably universal, ultrahomogeneous coherence spaces.*

Proof. Enumerate the two spaces. Starting from the empty partial isomorphism, alternate forth and back steps. In each step the extension property supplies a fresh event realizing the finite coherence pattern prescribed by the event being added. The union of the resulting chain is an isomorphism. The final statement follows from Proposition 4.5, since countable universality implies finite universality. □

5. Comparison of the two universal structures

Both C_ω and R_{coh} contain an induced copy of every at-most-countable coherence space, but the proofs reveal different internal behaviour. The construction of C_ω stores a finite relation code in a partial function. Its empty code is a definable exceptional event. The construction of R_{coh} instead realizes every finite extension pattern, and this property is precisely what drives both countable universality and ultrahomogeneity.

The universal structures also agree at the level relevant to linear retractions. By Proposition 2.4, every state domain $D(E)$ is a linear retract of both $D(C_\omega)$ and $D(R_{\mathrm{coh}})$. Homogeneity is additional symmetry of the ambient coherence space; it is not needed merely to obtain the retraction.

The comparison has been formalized in Lean 4. The development defines the two coherence graphs, verifies the exceptional empty event of C_ω, and formalizes the classical Rado extension-property argument together with its universality, ultrahomogeneity, and uniqueness consequences. It also checks the section–retraction and idempotent equations used below. Formalization is not a claim that the classical Rado results are new, and the development does not reprove the complete categorical coherence theory of every linear-logic connective.

Appendix A. Coherence-space semantics of propositional linear logic

This appendix recalls the standard coherence-space interpretation needed for the universal-retract application; see Girard [4, 5] and Girard, Lafont, and Taylor [6]. It also fixes the precise convention for the idempotent, set-based exponential. The categorical constructions and the soundness theorem in this appendix are mathematical background: the accompanying Lean development does not formalize them in full.

A.1. Formulae and linear negation

Definition A.1. Fix positive atomic propositions $p, q, r, \ldots$. Formulae of propositional classical linear logic are generated by

$$A, B ::= p \mid p^{\perp} \mid \mathbf{1} \mid \bot \mid \top \mid \mathbf{0} \mid A \otimes B \mid A \parr B \mid A \,\&\, B \mid A \oplus B \mid !A \mid ?A.$$

Linear implication is the abbreviation

$$A \multimap B = A^{\perp} \parr B.$$

Linear negation is the involution determined by

$$(p^{\perp})^{\perp} = p, \quad \mathbf{1}^{\perp} = \bot, \quad \bot^{\perp} = \mathbf{1}, \quad \top^{\perp} = \mathbf{0}, \quad \mathbf{0}^{\perp} = \top,$$

$$(A \otimes B)^{\perp} = A^{\perp} \parr B^{\perp}, \qquad (A \parr B)^{\perp} = A^{\perp} \otimes B^{\perp},$$

$$(A \,\&\, B)^{\perp} = A^{\perp} \oplus B^{\perp}, \qquad (A \oplus B)^{\perp} = A^{\perp} \,\&\, B^{\perp},$$

and $(!A)^{\perp} = ?A^{\perp}$, $(?A)^{\perp} = !A^{\perp}$.

A one-sided sequent $\vdash A_1, \ldots, A_n$ is a finite multiset of formulae. Exchange is therefore implicit. The usual axiom, cut, multiplicative, additive, weakening, contraction, dereliction, and promotion rules are interpreted below.

A.2. Linear traces and multiplicatives

Definition A.2. For coherence spaces E and F, define the dual by $|E^{\perp}| = |E|$ and, for distinct events,

$$x \smile_{E^{\perp}} y \quad \Longleftrightarrow \quad x \not\smile_{E} y.$$

The tensor product has web $|E| \times |F|$ and relation

$$(e, f) \smile_{E \otimes F} (e', f') \quad \Longleftrightarrow \quad e \smile_{E} e' \text{ and } f \smile_{F} f'.$$

Put

$$E \parr F = (E^{\perp} \otimes F^{\perp})^{\perp}, \qquad E \multimap F = E^{\perp} \parr F.$$

The tensor unit $\mathbf{1}$ is the one-event coherence space and $\bot = \mathbf{1}^{\perp}$ is the par unit.

The web of $E \multimap F$ is $|E| \times |F|$. Its relation can be read directly: two events (e, f) and (e', f') are coherent precisely when they are equal, or $e \not\smile_{E} e'$, or $f \neq f'$ and $f \smile_{F} f'$. This is the condition that makes states of $E \multimap F$ deterministic linear traces.

Definition A.3. A *linear trace* $R : E \to F$ is a state

$$R \in D(E \multimap F).$$

For $R : E \to F$ and $S : F \to G$, composition is relational composition,

$$S \circ R = \{(e, g) : \exists f \, ((e, f) \in R \text{ and } (f, g) \in S)\},$$

and the identity is $\{(e, e) : e \in |E|\}$.

Proposition A.4. *Linear traces are closed under composition. Composition is associative and the diagonal traces are identities.*

Proof. Suppose $(e, g), (e', g') \in S \circ R$ have witnesses f, f' and $e \smile_E e'$. If $g \neq g'$, the trace conditions for R and S imply $g \smile_G g'$. If $g = g'$ but $e \neq e'$, then either $f = f'$, contradicting the trace condition for R, or $f \neq f'$, in which case R makes $f \smile_F f'$ and the two occurrences of the same output in S give a contradiction. Thus $S \circ R$ is a state of $E \multimap G$. Associativity and the identity laws are the corresponding laws for relations. □

Every trace $R : E \to F$ induces a state map

$$\widehat{R} : D(E) \longrightarrow D(F), \qquad \widehat{R}(X) = \{f : \exists e \in X \ (e, f) \in R\}.$$

If $f \in \widehat{R}(X)$, its input witness in X is unique: two distinct coherent events cannot be related by a trace to the same output event.

Definition A.5. A map $h : D(E) \to D(F)$ is *linear stable* when it is Scott-continuous, $h(\varnothing) = \varnothing$, it preserves every bounded binary union, and each $f \in h(X)$ has a least input substate witnessing its membership, which is a singleton. Write $[D(E) \to_{\text{lin}} D(F)]$ for these maps.

Theorem A.6. *There is a natural bijection*

$$D(E \multimap F) \cong [D(E) \to_{\text{lin}} D(F)].$$

It sends R to $\widehat{R}$. Conversely, it sends h to

$$\mathrm{Tr}(h) = \{(e, f) : f \in h(\{e\})\}.$$

Proof. The trace condition makes $\widehat{R}(X)$ a state. Direct images commute with directed unions, preserve the empty state and bounded unions, and the preceding uniqueness argument gives the singleton witness.

Conversely, let $(e, f), (e', f') \in \mathrm{Tr}(h)$ with $e \smile_E e'$. Bounded-union preservation puts f and f' in the state $h(\{e, e'\})$. If $f = f'$ and $e \neq e'$, the same output is witnessed by both singletons, contradicting the existence of a least singleton witness: it would have to lie in $\{e\} \cap \{e'\} = \varnothing$, whereas $h(\varnothing) = \varnothing$. Hence $\mathrm{Tr}(h)$ is a state of $E \multimap F$. On finite states, bounded-union preservation gives

$$h(a) = \bigcup_{e \in a} h(\{e\});$$

Scott continuity extends this equality to all states. The two constructions are therefore inverse. □

Theorem A.7. *Coherence spaces and linear traces form a symmetric monoidal closed category, with tensor $\otimes$, unit $\mathbf{1}$, and internal hom $\multimap$. Explicitly,*

$$\mathbf{Coh}(E \otimes F, G) \cong \mathbf{Coh}(E, F \multimap G).$$

Proof. Currying sends a trace $R : (E \otimes F) \to G$ to

$$\Lambda R = \{(e, (f, g)) : ((e, f), g) \in R\}.$$

The explicit coherence relation in the linear implication shows that this is a trace $E \to (F \multimap G)$. Reassociating the triples gives the inverse. The symmetry, associativity, and unit maps are induced by the corresponding bijections of cartesian products. □

A.3. Additives and exponentials

Definition A.8. On the disjoint union $|E| \sqcup |F|$, the additive product $E \,\&\, F$ retains coherence within each summand and declares every cross-summand pair coherent. The additive coproduct $E \oplus F$ retains coherence within each summand and declares every cross-summand pair incoherent. The empty coherence space interprets both $\top$ and $\mathbf{0}$ in this basic model.

The usual projections and pairings make $E \,\&\, F$ a categorical product; the summand embeddings and copairings make $E \oplus F$ a categorical coproduct. In particular,

$$D(E \,\&\, F) \cong D(E) \times D(F).$$

Definition A.9. The events of $!E$ are the finite states of E:

$$|!E| = D_{\mathrm{fin}}(E), \qquad a \smile_{!E} b \iff a \cup b \in D(E).$$

Put $?E = (!(E^{\perp}))^{\perp}$. For a trace $R : E \to F$, declare $(a, b) \in !R$ when there is a finite subrelation $\rho \subseteq R$ whose first and second projections are a and b, respectively.

The structural traces are

$$\mathrm{der}_E = \{(\{e\}, e) : e \in |E|\}, \qquad \mathrm{weak}_E = \{(\varnothing, *)\} : !E \to \mathbf{1},$$

$$(a, (b, c)) \in \mathrm{contr}_E \iff a = b \cup c,$$

and

$$(a, \mathcal{A}) \in \mathrm{dig}_E \iff \mathcal{A} \in D_{\mathrm{fin}}(!E) \text{ and } \bigcup \mathcal{A} = a.$$

Theorem A.10. *The operation* $!$ *is a functor. Together with* der *and* dig *it is a comonad, while* weak *and* contr *make* $!E$ *a natural commutative comonoid. Moreover,*

$$!(E \,\&\, F) \cong !E \otimes !F, \qquad !\top \cong \mathbf{1}.$$

Consequently these coherence spaces interpret the exponential rules of propositional linear logic.

Proof. If $a \cup a'$ is a state and $(a, b), (a', b') \in !R$, the trace condition for R makes $b \cup b'$ a state; if $b = b'$, uniqueness of input witnesses gives $a = a'$. Hence $!R$ is a trace. Identity and composition follow by composing finite witnessing subrelations. The comonad and comonoid equations reduce to the unit, associativity, commutativity, and flattening laws for finite unions. A finite state of $E \,\&\, F$ decomposes uniquely into its two summand components, giving the first displayed isomorphism; the second follows because the empty coherence space has exactly one finite state. This is the standard idempotent coherence-space exponential [6, 9]. □

A.4. Interpretation of formulae, sequents, and proofs

Definition A.11. A valuation assigns an at-most-countable coherence space $v(p)$ to every positive atom. Define

$$[\![p]\!] = v(p), \qquad [\![p^{\perp}]\!] = v(p)^{\perp},$$

interpret the four binary connectives and the exponentials by the operations above, and put

$$[\![\mathbf{1}]\!] = \mathbf{1}, \quad [\![\perp]\!] = \mathbf{1}^{\perp}, \quad [\![\top]\!] = \varnothing, \quad [\![\mathbf{0}]\!] = \varnothing.$$

Then $[\![A^{\perp}]\!] \cong [\![A]\!]^{\perp}$. For a sequent $\Gamma = A_1, \ldots, A_n$, set

$$[\![\Gamma]\!] = [\![A_1]\!] \parr \cdots \parr [\![A_n]\!],$$

with the empty context interpreted by the par unit. A proof of $\vdash \Gamma$ is interpreted as a state of $[\![\Gamma]\!]$, equivalently a trace $\mathbf{1} \to [\![\Gamma]\!]$.

The axiom $\vdash A, A^{\perp}$ is interpreted by the identity trace on $[\![A]\!]$, and cut is interpreted by trace composition. The tensor rule uses the tensor functor, while the par rule uses the associativity isomorphism. The with-rule is product pairing, and the two plus-rules are the coproduct injections. Weakening, contraction, and dereliction for $?A$ are dual to the structural traces of $!A$. Promotion is coKleisli lifting: if $f : !X \to [\![A]\!]$ interprets its premise, then its conclusion is interpreted by

$$f^{\dagger} = !f \circ \mathrm{dig}_X : !X \longrightarrow ![\![A]\!].$$

Theorem A.12. *For every valuation and every derivation* π *of* $\vdash \Gamma$*, the recursive interpretation above gives a well-defined state*

$$[\![\pi]\!] \in D([\![\Gamma]\!]).$$

The standard cut-elimination equations represented by the symmetric monoidal, product, coproduct, comonad, and comonoid laws preserve denotation.

Proof. Induct on the derivation. Axiom and cut use identities and composition of traces. The multiplicative cases use symmetric monoidal closure, the additive cases use the product and coproduct universal properties, and the exponential cases use Theorem A.10. The corresponding cut-elimination steps are exactly the associativity, unit, naturality, product/coproduct, and comonad/comonoid equations used by this interpretation. □

At the domain level, a typed denotation $A \to B$ is therefore a linear stable map

$$D(\llbracket A \rrbracket) \longrightarrow D(\llbracket B \rrbracket),$$

and Theorem A.6 gives

$$D(\llbracket A \multimap B \rrbracket) \cong [D(\llbracket A \rrbracket) \to_{\mathrm{lin}} D(\llbracket B \rrbracket)].$$

This establishes a sound interpretation. Since $\top$ and $\mathbf{0}$ are both interpreted by the empty coherence space, no formula-separating completeness statement is asserted.

Appendix B. The universal retract presentation

Fix an at-most-countable coherence-space valuation and put $U = D(R_{\mathrm{coh}})$. The webs used to interpret formulae remain at most countable: duality does not change a web, finite products and disjoint unions preserve countability, and the set of finite subsets of a countable set is countable. Thus every $\llbracket A \rrbracket$ admits an induced embedding into R_{coh}.

Theorem B.1. *Let $U = D(R_{\mathrm{coh}})$. For every formula A there are linear stable maps*

$$i_A : D(\llbracket A \rrbracket) \longrightarrow U, \qquad p_A : U \longrightarrow D(\llbracket A \rrbracket)$$

such that $p_A i_A = \mathrm{id}$. Hence

$$e_A = i_A p_A : U \longrightarrow U$$

is a linear idempotent whose image is isomorphic to $D(\llbracket A \rrbracket)$.

Proof. Corollary 4.6 supplies an induced embedding $\varphi_A : \llbracket A \rrbracket \hookrightarrow R_{\mathrm{coh}}$. Apply Proposition 2.4 and set $i_A = I_{\varphi_A}$ and $p_A = P_{\varphi_A}$. Then

$$e_A^2 = i_A(p_A i_A)p_A = i_A p_A = e_A.$$

□

Definition B.2. Let $\mathrm{End}_{\mathrm{lin}}(U)$ be the monoid of linear stable endomaps of U. The Karoubi envelope generated by U has as objects the idempotents $e : U \to U$. A morphism from e to d is an endomap $g : U \to U$ satisfying

$$g = dge.$$

Composition is ordinary composition, and the identity on the object e is e itself. Denote this category by $\mathrm{Kar}(U)$.

Theorem B.3. *Let*

$$f : D(\llbracket A \rrbracket) \longrightarrow D(\llbracket B \rrbracket)$$

be a linear stable map and define $\widehat{f} = i_B f p_A : U \to U$. Then

$$\widehat{f} = e_B \widehat{f} e_A.$$

Conversely, every linear endomap $g : U \to U$ satisfying $g = e_B g e_A$ determines the typed map $p_B g i_A$. These two assignments are inverse, and hence

$$[D(\llbracket A \rrbracket) \to_{\mathrm{lin}} D(\llbracket B \rrbracket)] \cong \mathrm{Kar}(U)(e_A, e_B).$$

Proof. Using $p_A i_A = \mathrm{id}$ and $p_B i_B = \mathrm{id}$,

$$e_B \widehat{f} e_A = (i_B p_B)(i_B f p_A)(i_A p_A) = i_B f p_A = \widehat{f}.$$

Returning to the typed map gives

$$p_B(i_B f p_A) i_A = f.$$

If $g = e_B g e_A$, encoding $p_B g i_A$ again gives

$$i_B(p_B g i_A) p_A = e_B g e_A = g.$$

□

Corollary B.4. *The endomap representation preserves identities and composition. The identity of $D(\llbracket A \rrbracket)$ is represented by e_A, and if $f : A \to B$ and $h : B \to C$ are typed denotations, then*

$$\widehat{h \circ f} = \widehat{h} \circ \widehat{f}.$$

Proof. The identity equation is $i_A \,\mathrm{id}\, p_A = i_A p_A = e_A$. For composition, use $p_B i_B = \mathrm{id}$:

$$(i_C h p_B)(i_B f p_A) = i_C h (p_B i_B) f p_A = i_C h f p_A.$$

□

Thus the countable coherence-space semantics may be presented inside the endomorphism monoid of the single state domain U: formulae correspond to linear idempotents, and typed denotations are the morphisms between those idempotents in the Karoubi envelope. The embeddings φ_A are choices, so this is a universal retract presentation rather than a canonical pointwise coding. The argument establishes a representation of the standard sound coherence-space semantics; it does not assert a new completeness theorem for linear logic.

Appendix C. The state domain of the coordinate universal space

The homogeneous space R_{coh} is the appropriate symmetric ambient object, whereas the finite-coordinate carrier C_ω retains a useful concrete description inherited from the 2015 construction.

Theorem C.1. *Let $\mathcal{P} = \mathrm{Fn}_{<\omega}(\mathbb{N}, 2)$. Then*

$$D(C_\omega) = \{X \subseteq \mathcal{P} : \forall s \neq t \in X,\ s \# t\}.$$

Thus its states are precisely the pairwise-conflicting families of finite partial binary functions, ordered by inclusion.

Proof. A subset $X \subseteq \mathcal{P}$ is a state exactly when $s \smile_\omega t$ for every $s, t \in X$. By Definition 3.1, distinct members are coherent exactly when they conflict. □

For $s \in \mathcal{P}$, let

$$[s] = \{\alpha \in 2^{\mathbb{N}} : s \subseteq \alpha\}$$

be the basic Cantor cylinder determined by s.

Proposition C.2. *For $s, t \in \mathcal{P}$,*

$$s \# t \quad \Longleftrightarrow \quad [s] \cap [t] = \varnothing.$$

Consequently, states of C_ω can equivalently be viewed as pairwise-disjoint families of basic Cantor cylinders.

Proof. If s and t conflict, no total binary sequence extends both. If they do not conflict, their union is a finite partial function and extends to a member of $2^{\mathbb{N}}$, so the two cylinders meet. □

Theorem C.3. *Let*

$$\mathcal{K}_\omega = \{a \subseteq_{\mathrm{fin}} \mathcal{P} : \forall s \neq t \in a,\ s \# t\},$$

ordered by inclusion. Then

$$D(C_\omega) \cong \mathrm{Idl}(\mathcal{K}_\omega).$$

The isomorphism sends a state X to

$$I_X = \{a \in \mathcal{K}_\omega : a \subseteq X\},$$

and an ideal I to $X_I = \bigcup I$.

Proof. For a state X, the set I_X is nonempty and downward closed. It is directed because the union of two finite substates of X is again a finite substate. Conversely, let I be an ideal. If distinct $s, t \in \bigcup I$, choose $a, b \in I$ containing them. Directedness gives $c \in I$ with $a \cup b \subseteq c$, and membership of c in $\mathcal{K}_\omega$ yields $s \# t$. Hence X_I is a state.

Clearly $X_{I_X} = X$. If $a \subseteq X_I$ is finite, directedness supplies a single $b \in I$ containing every member of a; downward closure then gives $a \in I$. Therefore $I_{X_I} = I$. □

This ideal-completion presentation identifies the compact elements with the finite states of C_ω. It explains why the coordinate object is useful for explicit coding even though its distinguished empty event prevents ultrahomogeneity.

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